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NMNH DNA Extraction Protocol using Qiagen DNeasy Kit

1 PROTOCOL INFORMATION

1.1 Minimum Information about an Omics Protocol (MIOP)

1.2 Making eDNA FAIR (FAIRe)

- FAIRe terms are listed in the YAML frontmatter of this page.
- See <https://fair-edna.github.io/download.html> for the FAIRe checklist and more information.
- See <https://fair-edna.github.io/guidelines.html#missing-values> for guidelines on missing values that can be used for missing FAIRe or MIOP terms.

1.3 Authors

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K. Murphy	Smithsonian Institution, Laboratories of Analytical Biology	Not available	2020-03-18

1.4 Related Protocols

1.4.1 Internal Protocols

PROTOCOL NAME	LINK	VERSION	RELEASE DATE
Content	Content	Content	Content

1.4.2 External Protocols

PROTOCOL NAME	LINK	ISSUER / AUTHOR	ACCESS DATE
Comparison of capture and storage methods for aqueous macrobial eDNA using an optimized extraction protocol: advantage of enclosed filter	https://doi.org/10.1111/2041-210X.12683	Spens et al. 2017	2024-03-01
Downstream transport and seasonal variation in freshwater pearl mussel (<i>Margaritifera margaritifera</i>) eDNA concentration	https://doi.org/10.1002/edn3.10	Wacker et al. 2019	2024-03-01
DNeasy Blood & Tissue Kit Handbook	https://www.qiagen.com/us/resources/resource-detail?id=6b09dfb8-6319-464d-996c-79e8c7045a50	QIAGEN	2024-03-01

1.5 Protocol Revision Record

VERSION	RELEASE DATE	DESCRIPTION OF REVISIONS
1.0.0	2020-03-18	Initial release (K. Murphy, Smithsonian Institution LAB)
1.1.0	2024-03-01	Adapted for OKEX project; updated authorship and affiliation
1.1.1	2026-04-10	Updated YAML front matter
1.1.2	2026-04-10	Updated YAML front matter with version number
1.1.3	2026-04-10	Updated README with Zenodo DOIs

1.6 Acronyms and Abbreviations

ACRONYM / ABBREVIATION	DEFINITION
eDNA	environmental DNA
NOAA	National Oceanic and Atmospheric Administration
NSL	National Systematics Laboratory

ACRONYM / ABBREVIATION	DEFINITION
SI	Smithsonian Institution
NMNH	National Museum of Natural History
LAB	Laboratories of Analytical Biology
ATL	Buffer ATL (QIAGEN lysis buffer)
AE	Buffer AE (QIAGEN elution buffer)
AL	Buffer AL (QIAGEN lysis buffer)
AW1	Buffer AW1 (QIAGEN wash buffer 1)
AW2	Buffer AW2 (QIAGEN wash buffer 2)
EtOH	Ethanol
ProtK	Proteinase K
RT	Room temperature

1.7 Glossary

SPECIALISED TERM	DEFINITION
Sterivex capsule	An enclosed inline filter capsule (0.22 µm or 0.45 µm) used to filter water samples for eDNA collection
Spin column	A silica membrane column used in kit-based DNA purification that retains DNA during centrifugation wash steps
Lysate	The liquid mixture containing cellular debris and DNA after enzymatic digestion
Eluate	The liquid collected after DNA is washed off a spin column membrane with elution buffer

2 BACKGROUND

2.1 Summary

This document describes the protocol to extract environmental DNA (eDNA) from seawater samples collected on sterivex capsule filters using the Qiagen DNeasy Blood & Tissue Kit. The protocol is adapted from Spens et al. (2017) and Wacker et al. (2019) and was updated on 18-Mar-2020 by K. Murphy (Smithsonian Institution, Laboratories of Analytical Biology). It is applied here to samples collected at sea on the NOAA Ship Okeanos Explorer.

2.2 Method Description and Rationale

This protocol provides a method to extract and purify environmental DNA from eDNA samples preserved in sterivex capsule filters filled with Buffer ATL (or stored dry). Enzymatic lysis using Proteinase K followed by silica membrane spin column purification (DNeasy kit) enables efficient recovery of high-quality DNA from filtered seawater samples.

2.3 Spatial Coverage and Environment(s) of Relevance

Oceanic Epipelagic Zone [ENVO:01000035] Marine Mesopelagic Zone [ENVO:00000213] Marine Bathypelagic Zone [ENVO:00000211] Marine Abyssopelagic Zone [ENVO:00000212]

3 PERSONNEL REQUIRED

This procedure can be performed by one individual trained in basic wet lab bench techniques.

3.1 Safety

Basic laboratory safety measures and appropriate personal protective equipment (nitrile gloves, lab coat, eye protection) should be used. Work should be conducted in a laminar flow hood when possible. Standard precautions for working with ethanol and bleach solutions apply.

3.2 Training Requirements

Basic wet lab bench skills including pipetting, aseptic technique, and centrifuge operation are required. Familiarity with spin column-based DNA extraction protocols is recommended.

3.3 Time Needed to Execute the Procedure

Total time (estimated): ~48 hours across 2 days - Day 1 sample preparation and incubation: 1–2 hours active time + 24 hours incubation - Day 2 extraction and elution: 3–4 hours active time

4 EQUIPMENT

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
Durable equipment				
Laminar flow hood	Nonspecific	Nonspecific	1	UV sterilize between uses
Benchtop vortexer	Nonspecific	Nonspecific	1	N/A
Heat block or incubator with shaking	Nonspecific	Nonspecific	1	Must maintain 56°C with agitation
Refrigerated centrifuge	Nonspecific	Nonspecific	1	Must reach 20,000g; compatible with 1.5/2.0 mL and 5.0 mL tubes
P1000 Pipette	Nonspecific	Nonspecific	1	Or similar
P200 Pipette	Nonspecific	Nonspecific	1	Or similar
P20 Pipette	Nonspecific	Nonspecific	1	Or similar
Consumable equipment				
3 mL syringes	Fisher 14-955-457	Fisher Scientific	1 per sample	For lysate recovery from sterivex inlet port
Outlet caps (Female Luer)	Cole-Parmer EW-30800-12	Cole-Parmer	1 per sample	Only needed if sterivex units are not already capped
Inlet caps (Male Luer)	Cole-Parmer EW-30800-30	Cole-Parmer	1 per sample	Only needed if sterivex units are not already capped
Eppendorf LoBind Tubes (5 mL)	Eppendorf 30108310	Eppendorf	At least 1 per sample	For lysate collection

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
Eppendorf LoBind Tubes (2 mL)	Eppendorf 22431048	Eppendorf	At least 2 per sample	For final eluate storage
DNeasy spin columns and collection tubes	Included in DNeasy Blood & Tissue Kit	QIAGEN	1 per sample	From kit
Parafilm	Nonspecific	Nonspecific	As needed	For sealing sterivex ports
Filter tips	Nonspecific	Nonspecific	As needed	Use for all steps
Hype-Wipes	Current Technologies 9103	Current Technologies	As needed	For surface decontamination
KimWipes	KimwipesTM 34155	Kimberly-Clark	1 box	As needed
Chemicals DNeasy Blood & Tissue Kit	QIAGEN 69504 or 69506	QIAGEN	As needed	Contains spin columns, collection tubes, and Buffers AW1, AW2, AE
Proteinase K	QIAGEN 19133	QIAGEN	As needed	Alternatively, prepare 20 mg/mL stock in-house
Buffer AL	QIAGEN 19075	QIAGEN	As needed	For lysis step; mix fresh 1:1 with 100% EtOH on day of extraction
Buffer ATL	QIAGEN 939011	QIAGEN	As needed	Only needed if sterivex units were stored dry (not pre-filled with ATL)
Absolute Ethanol (100%)	Nonspecific	Nonspecific	As needed	Mixed with Buffer AL; do not store EtOH:AL mixture
Ethanol (70%)	Nonspecific	Nonspecific	As needed	For surface decontamination
10% Bleach or Hype-Wipes	Nonspecific	Nonspecific	As needed	For surface decontamination
Surface Disinfectant	DNA Away	Thermo-Fisher	1 bottle	For decontamination of pipettes and surfaces

5 STANDARD OPERATING PROCEDURE

5.1 Methodological Notes

- All eDNA extractions should ideally be performed in a laminar flow hood.
- The hood, any necessary bench space, and pipettes must be cleaned with 10% bleach and 70% ethanol

immediately prior to use. The hood should be UV sterilized between uses.

- Before starting, calculate how much Proteinase K, Buffer ATL, and Buffer AL will be needed and confirm sufficient supply before beginning extractions.
- Prepare a fresh EtOH:AL mixture on the day of extraction; do not store this mixture.
- Use filter tips for all extractions.
- Always include an extraction blank alongside sample preparations.

5.2 Preparation

1. Decontaminate and disinfect the work area by wiping down the work surface with 10% bleach or a Hype-Wipe, followed by 70% ethanol. Pipettes should be wiped down with DNA Away followed by 70% ethanol, then UV-irradiated for 15 minutes in the fume hood or laminar flow cabinet.
2. Remove sterivex capsules from storage. If samples were stored at -80°C , thaw to room temperature. Wipe the outside of each sterivex capsule with 10% bleach or a Hype-Wipe and allow to dry, then wipe with 70% ethanol and allow to dry.
3. Ensure that the outlet cap is firmly in place and para-filmed to prevent leakage.
 - If the outlet was previously covered only with parafilm, add an outlet cap before proceeding. Remove the original parafilm, add an outlet cap, and re-parafilm.
4. An extraction negative control (extraction blank) should be prepared at this time. Use a sterivex capsule containing only Buffer ATL with no sample filtrate, or follow your laboratory's standard blank protocol.

5.3 Day 1 — Lysis

5. Carefully remove the inlet cap from each sterivex unit.
6. ONLY if samples were stored dry (not pre-filled with ATL): add 1.5 mL Buffer ATL through the inlet port. Omit this step if samples already contain ATL.
7. Add 150 μL of Proteinase K through the inlet port.
 - Note: if a volume other than 1.5 mL was used for ATL, use 1/10th that volume for Proteinase K (e.g., 2 mL ATL $\rightarrow$ 200 μL Proteinase K).
8. Cap the inlet port with an inlet cap and seal with parafilm to prevent leakage. Do not rely on parafilm alone — use an inlet cap.
9. Invert 5 times to mix.
10. Incubate at 56°C for 24 hours with shaking/agitation.
 - If incubation is less than 24 hours (e.g., overnight), record the actual length of incubation.

5.4 Day 2 — Extraction

11. Preheat Buffer AE to 56°C in a heat block.
12. Remove samples from incubator. Shake each capsule vigorously by hand five times.
13. Using a 3 mL syringe attached to the inlet port, remove all liquid from the sterivex capsule. Record the volume and transfer the lysate to a labeled 5 mL Lo-Bind tube.
14. Prepare the EtOH:AL mixture fresh: mix 100% EtOH and Buffer AL in a 1:1 ratio, preparing sufficient volume to add 2 volumes of EtOH:AL mixture per 1 volume of lysate across all samples.
 - Example: 1.5 mL of lysate requires 3.0 mL of EtOH:AL mixture.
15. Add 2 volumes of EtOH:AL mixture to each tube of lysate. Vortex vigorously to mix.

16. Pipette 650 μ L of the mixture into a DNeasy spin column seated in a 2 mL collection tube (from kit).
17. Centrifuge at 6,000g (8,000 rpm) for 1 minute, then discard flow-through.
18. Repeat steps 16–17 until the entire sample volume has been passed through the same spin column, discarding the flow-through each time.
19. Place the spin column into a new 2 mL collection tube (from kit).
20. Add 500 μ L Buffer AW1, then centrifuge for 1 minute at 6,000g (8,000 rpm). Discard flow-through and collection tube.
21. Place the spin column into a new 2 mL collection tube. Add 500 μ L Buffer AW2, then centrifuge for 3 minutes at 20,000g (14,000 rpm). Discard flow-through.
22. Replace the spin column onto the same collection tube and centrifuge again for 1 minute at 17,000g (13,000 rpm) to dry the membrane.
23. Transfer the spin column to a new, labeled 2 mL DNA Lo-Bind tube with the cap removed.
24. Add 100 μ L of 56°C Buffer AE directly to the filter membrane.
25. Incubate at room temperature for 10 minutes.
26. Centrifuge for 1 minute at 6,000g (8,000 rpm).
27. Add the eluate back onto the spin column membrane, then incubate at room temperature for another 10 minutes.
28. Centrifuge for 1 minute at 6,000g (8,000 rpm).
29. Discard the spin column.
30. Transfer the DNA eluate to a new, labeled 2.0 mL DNA Lo-Bind tube.
31. Optional: Aliquot 3–5 μ L to a separate tube for DNA quantification.
32. Store DNA at -20°C.

5.5 Quality Control

No quantification step is specified in the source protocol beyond optional aliquoting for downstream quantification (step 31). If quantification is desired, a Qubit dsDNA HS Assay or equivalent fluorometric method is recommended. Record DNA concentration [ng/ μ L] for each sample.

5.6 Basic Troubleshooting Guide

PROBLEM	POSSIBLE CAUSE	SUGGESTED ACTION
Low DNA yield	Insufficient lysis time or ATL volume	Confirm 24-hour incubation; confirm ATL was added if stored dry
Low DNA yield	Incomplete lysate recovery from sterivex	Use syringe carefully to withdraw all liquid from capsule
Contamination in extraction blank	Cross-contamination during pipetting	Re-decontaminate workspace; replace pipette tips; UV sterilize hood
EtOH:AL mixture precipitate	Old or improperly mixed reagents	Prepare fresh EtOH:AL mixture on day of extraction; do not store

PROBLEM	POSSIBLE CAUSE	SUGGESTED ACTION
Leakage from sterivex during incubation	Inadequate sealing of ports	Confirm outlet and inlet caps are in place and parafilmed before incubation

6 REFERENCES

Spens, J., Evans, A. R., Halfmaerten, D., Knudsen, S. W., Sengupta, M. E., Mak, S. S., Sigsgaard, E. E., and Hellström, M. (2017). Comparison of capture and storage methods for aqueous microbial eDNA using an optimized extraction protocol: advantage of enclosed filter. *Methods in Ecology and Evolution*, 8: 635–645. <https://doi.org/10.1111/2041-210X.12683>

Wacker, S., Fossøy, F., Larsen, B. M., Brandsegg, H., Sivertsgård, R., and Karlsson, S. (2019). Downstream transport and seasonal variation in freshwater pearl mussel (*Margaritifera margaritifera*) eDNA concentration. *Environmental DNA*, 1: 64–73. <https://doi.org/10.1002/edn3.10>

7 APPENDIX A: DATASHEETS

Not applicable in this version.